

Scott Novakowski

Senior Software Engineer

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[GitHub](#) | [Website](#) | [LinkedIn](#)

SUMMARY

My diverse work spans 3D graphics, immersive VR/AR, video encoding, real-time multiplayer, and mobile applications. I'm driven by genuine curiosity and a desire to delve deeply into problems, which makes me particularly drawn to projects that blend technical complexity with creative potential. I thrive when my work has a meaningful impact.

SKILLS

- **Languages:** JavaScript, TypeScript, C, Objective-C, Swift, PHP, SQL, GLSL
- **3D Graphics & Immersive:** OpenGL, OpenGL ES2, WebGL, Three.js, WebXR, 360° and Stereoscopic Video, Virtual Reality (VR), Augmented Reality (AR)
- **Frameworks:** Three.js, React, Vite, Webpack
- **Platforms:** Web, macOS, iOS (iPhone), tvOS (Apple TV), Meta Quest (VR)
- **Backend & Realtime:** MySQL, WebRTC, PeerJS, Ant Media Server (Multi-Client Video Streaming), Colyseus (Multiplayer Framework), FeathersJS (Real-time Framework)
- **Leadership:** Team leadership, intern supervision, student mentorship

EXPERIENCE

Senior Developer

Liquid Cinema — November 2015 to Present

- <https://liquidcinemavr.com>
- Built the frontend web component of the Liquid Cinema platform, a browser-native, interactive cinematic 360° video WebGL player
- Migrated the web player from a raw WebGL app built on an internal rendering library I developed to Three.js
- Developed backend storage for user viewing directions and a frontend heat map-style visualization to represent view concentration
- Developed client-side Apple iOS and tvOS components of the Liquid Cinema platform
- Developed virtual and augmented reality support using the WebXR standard for the web-

based Liquid Cinema platform

- FFmpeg video encoding and Vimeo video hosting integration
- OMAF 360° VR Video Streaming
 - Fraunhofer HHI collaboration to integrate OMAF 360° VR video streaming into the Liquid Cinema web platform
 - <https://www.hhi.fraunhofer.de/en/departments/vca/technologies-and-solutions/mpeg-omaf.html>
- "Magic of Flight" interactive VR educational experience, Lead WebXR Developer
 - Meta collaboration to create an interactive web experience for the Quest VR Headset
 - 2021 WebXR Poly Awards winner:
 - Education Experience of the Year
 - Video Experience of the Year
 - Experience of the Year
 - <https://liquidcinemavr.com/fly/>
- Multi-user WebRTC live web video chat integration using PeerJS with Colyseus and Feathers backend systems
- Senior frontend lead for a collaborative metaverse environment
 - Conestoga College collaboration
 - 3D avatar conferencing solution
 - Virtual reality multi-user live interactive environment
 - Messaging and live video and screen-share streaming

The Lindsay Project: Lead Programmer and Software Designer

University of Calgary — 2009 to 2015

- Funded by the department of Undergraduate Medical Education in the Cumming School of Medicine at the University of Calgary
- Worked with Zygote 3D human anatomy models, originally used in Google Body, now Zygote Body, as part of a full stack pipeline for multiple applications
- Developed "Zygote 3D Anatomy Atlas & Dissection Lab"
 - Released as an iPhone and iPad app on the iTunes App Store

- Now archived, references available:
 - <https://lindsayvirtualhuman.com/?p=273>
 - https://www.youtube.com/watch?v=3MZps2_Z1zo
 - <https://www.youtube.com/watch?v=38d7P3JB4SE>
 - https://www.youtube.com/watch?v=M0xubQ0_5Q0
- Developed "Atlas", a 3D human anatomy education-oriented web application
 - CBC segment on The Lindsay Project, featuring the Atlas application:
 - <https://www.cbc.ca/news/canada/calgary/new-medical-tool-honours-u-of-c-student-s-memory-1.1240316>
- Developed "Presenter", a Mac application for creating 3D human anatomy slide presentations.
 - https://lindsayvirtualhuman.com/?page_id=451

Vaccine Design and Implementation Project

University of Calgary — 2008

- Developed interactive real-time 3D OpenGL visualizations for a vaccine design project funded through the AHFMR Interdisciplinary Team in Vaccine Design and Implementation program

Swarm Art Software Developer

University of Calgary — 2005 to 2007

- Built agent-based swarm systems to generate visuals and facilitate interactive art displays
- Featured in:
 - Images published in the Leonardo Journal, Volume 40, issue 3, by MIT Press, 2007
 - Digital'06: "Bio/Med SciART" submission accepted for public display in the New York Hall of Science, 2006
 - Victoria Conference Center street front window interactive public display, Victoria BC, August to December 2006
 - Discovery Channel "A Daily Planet" segment on swarm intelligence, featuring the Swarm Art system, aired on October 23, 2006

- Nickle Galleries Museum interactive display, Calgary AB, 2005

EDUCATION

Bachelor of Science in Computer Science

University of Calgary, AB — 2006

PUBLICATIONS

- **LINDSAY Virtual Human: Multi-Scale, Agent-based, and Interactive**

C. Jacob, S. von Mammen, T. Davison, A. Sarraf-Shirazi, V. Sarpe, A. Esmaceli, D. Phillips, I. Yazdanbod, **S. Novakowski**, S. Steil, C. Gingras, H. Jamniczky, B. Hallgrimsson, and B. Wright. Advances in Intelligent Modelling and Simulation, Studies in Computational Intelligence Volume 422, pp 327-349; Springer-Verlag Berlin Heidelberg, 2012

- **Evolutionary Swarm Design: How Can Swarm-based Systems Help to Generate and Evaluate Designs?**

Sebastian von Mammen, **Scott Novakowski**, Gerald Hushlak, and Christian Jacob.

Design Principles and Practices: An International Journal, Volume 3, Issue 3, pp. 371-386, 2009.

- **Evolutionary design of dynamic SwarmScapes**

Namrata Khemka, **Scott Novakowski**, Gerald Hushlak, and Christian Jacob.

Genetic and Evolutionary Computation Conference, GECCO 2008, Proceedings, Atlanta, GA, USA, July 12-16, 2008.

- **Motion swarms: video interaction for art in complex environments**

Quoc Nguyen, **Scott Novakowski**, Jeffrey E. Boyd, Christian Jacob, and Gerald Hushlak.

Proceedings of the 14th annual ACM international conference on Multimedia, Santa Barbara, CA, USA, October 23-27, 2006.